

Please check the examination details below before entering your candidate information

Candidate surname

Other names

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Centre Number

Candidate Number

Pearson Edexcel International GCSE (9–1)

By Enidu Batuwanthudawe

Afternoon (Time: 1 hour 30 minutes)

Paper
reference

4IT1/01

Information and Communication

Technology (ICT)

PAPER 1: Written Paper

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, center number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
 - *there may be more space than you need.*

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
 - *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Yash hosts a live interactive radio stream using professional audio equipment



(Source:CircleBit.net)

1 Yash streams an online live radio show.

- (a) State one **specialized input device** (other than a basic microphone) that could be used to capture Yash's voice, and give one reason why it is suitable. (2)

- (b) Yash needs to minimize latency and ensure high-quality audio transmission between his input device and computer.

- (i) Which one of these is the most suitable connectivity method for this scenario? (1)

- ☐ A Bluetooth
- ☐ B NFC
- ☐ C USB (wired)
- ☐ D Wi-Fi

- (ii) Explain why your chosen answer in (b)(i) is more suitable than one other option. (2)

- (iii) Which one of these best describes the impact of high latency on a live stream?

- ☐ A Audio becomes delayed and out of sync with real-time events
- ☐ B Audio file size increases
- ☐ C Volume levels automatically increase
- ☐ D Sampling rate decreases

- (c) Describe in detail how streaming works when Yash broadcasts her radio show. (3)

(d) Yash wants to stream her show globally and allow listeners to join live.
State the type of network used and explain why it is required instead of a LAN (2)

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(e) Explain one way copyright legislation affects how a listener can use a recording of Yash’s live radio stream. (2)

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(f) **Yash’s computer has 16 GiB of RAM.**
Each uncompressed recording uses **512 MiB of RAM.**

(i) Construct an **expression** to show how many recordings can be stored in RAM at once. (2)

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(ii) Which one of these is a benefit to Yash of increasing RAM? (1)

- ☐ **A** More recordings can be processed simultaneously without slowing down
- ☐ **B** The internet speed will increase automatically
- ☐ **C** Audio files will become compressed automatically
- ☐ **D** The operating system will require less storage space

(iii) State two benefits of using secondary storage, rather than RAM, to save the radio recordings. (2)

1).....

2).....

(iv) Which one of these does the 'A' stand for in RAM?

(1)

- ☐ **A** Access
- ☐ **B** Allocation
- ☐ **C** Array
- ☐ **D** Available

(v) State the reason why RAM is described as 'random access'.

(2)

(g) **The audio data is processed.**

(i) State how the speed of a processor is measured.

(1)

(i) State one feature of a laptop that allows expansion or improvement of audio processing performance.

(1)

(h) **Listeners can subscribe to receive updates about new shows.**

(i) State one risk to a listener's personal information when subscribing online.

(1)

(ii) Describe one way a listener can protect their personal information when subscribing.

(2)

(iii) There are legal requirements for storing personal data.

Explain two legal requirements Yash must follow when storing listeners' personal information.

(4)

1).....